



REVIEW ARTICLE

THE DYNAMIC BINDING HYPOTHESIS: A NOVEL FRAMEWORK FOR UNDERSTANDING CONSCIOUSNESS

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Abstract

The Dynamic Binding Hypothesis (DBH) offers a framework for understanding consciousness as a product of synchronized neural oscillations across brain regions. This integration of brain activity, modulated by attention, working memory, and social context, is proposed as the basis of conscious experience. DBH suggests that disruptions in these synchronized oscillations may underlie consciousness disorders. Personalized medicine can enhance DBH's application by tailoring treatments based on individual neural characteristics. By using genetic data, biomarkers, and advanced diagnostics, therapies for consciousness disorders can be targeted to individual variations in neural oscillations. Furthermore, DBH and quantum biology offer complementary views. Quantum biology explores how quantum phenomena might influence neural activity, suggesting that quantum effects in microtubules and photons in neural communication could contribute to the neural synchronization central to DBH and consciousness. While insightful, DBH is not without limitations and should be considered alongside other theories to achieve a comprehensive understanding of consciousness. Integrating insights from various perspectives may be the most effective approach to unraveling this complex phenomenon.

Keywords: Consciousness disorders, dynamic binding hypothesis, neural oscillations, personalized medicine, quantum biology, working memory.

INTRODUCTION

Consciousness, the subjective experience of the world, remains one of the most profound mysteries of neuroscience. While significant progress has been made in understanding the neural correlates of consciousness, a comprehensive theory that can account for its complexity and richness has yet to emerge. Traditional theories, such as Global Workspace Theory and Integrated Information Theory, have provided valuable insights but often fall short in fully explaining the dynamic and subjective nature of conscious experience¹.

The Dynamic Binding Hypothesis (DBH) offers a novel framework that addresses these limitations. By emphasizing the role of neural oscillations, attention, working memory and social context, the DBH provides a more comprehensive and dynamic account of consciousness. This paper explores the key tenets of the DBH, its potential implications for understanding consciousness disorders, and future directions for research^{2,3}. Table 1 summarizes progressions made in the field by providing list of some research and findings⁴⁻²⁸.

The Dynamic Binding Hypothesis (DBH)

The DBH posits that consciousness arises from the dynamic integration of neural activity across multiple brain regions through synchronized oscillations. Neural oscillations, rhythmic patterns of electrical activity, provide a temporal framework for binding together information from different sensory modalities and cognitive processes^{29,30}. This integration process, facilitated by attention and working memory, gives rise to a unified conscious experience.

Key Tenets of the DBH

Neural oscillations: The synchronization of neural oscillations in different frequency bands (e.g., alpha, beta, and gamma) underlies the binding of information into coherent perceptual and cognitive representations. Studies have shown that gamma oscillations (30-100 Hz) are particularly important for the integration of sensory information and the formation of coherent perceptual experiences. For instance, Fries (2005) highlighted the role of gamma oscillations in visual perception, suggesting that these oscillations facilitate the binding of visual features into a unified object representation⁴.

Table 1: Some key research and progress made in the field of neural oscillations and consciousness⁴⁻²⁸.

Research/Progress	Key findings and summary
Neuronal Synchrony	Proposed that neuronal synchrony is a versatile code for defining relations in the brain.
Dynamic Predictions	Discussed the role of oscillations and synchrony in top-down processing.
Brainweb	Introduced the concept of phase synchronization and large-scale integration in the brain.
Neuronal Oscillations	Reviewed the role of neuronal oscillations in cortical networks.
Cognitive Dynamics	Proposed that neuronal communication through coherence is a mechanism for cognitive dynamics.
Rhythms of the Brain	Comprehensive review of brain rhythms and their functions.
Attention and Oscillations	Demonstrated that attention modulates the synchronization of beta and gamma oscillations.
Gamma Oscillations	Reviewed the association of human gamma-frequency oscillations with attention and memory.
Vegetative State Connectivity	Found that functional connectivity in the default network is preserved in vegetative state patients.
Social Interaction and Brain	Explored the brain basis of human social interaction using brain imaging.
Cortical Rhythms	Reviewed neurophysiological and computational principles of cortical rhythms in cognition.
Schizophrenia and Oscillations	Discussed abnormal neural oscillations and synchrony in schizophrenia.
Resting-State Activity	Emerging concepts for the dynamical organization of resting-state activity in the brain.
Phase Synchronization in Memory	Reviewed the role of phase synchronization in memory processes.
Large-Scale Interactions	Discussed spectral fingerprints of large-scale neuronal interactions.
Alpha-Band Oscillations	Reviewed alpha-band oscillations, attention and controlled access to stored information.
Theta-Gamma Neural Code	Proposed the interaction between theta and gamma oscillations supports working memory.
Rhythms for Cognition and Communication Through Coherence	Discussed communication through coherence as rhythms for cognition.
Cross-Frequency Coupling	Reviewed the functional role of cross-frequency coupling.
Phase-Dependent Coding	Investigated phase-dependent neuronal coding of objects in short-term memory.
TMS-EEG Studies	Provided new insights into rhythmic brain activity from TMS-EEG studies.
Neural Synchrony Development	Discussed neural synchrony and the development of cortical networks.
Alpha-Frequency Oscillations	Explored new vistas for alpha-frequency band oscillations.
Cognitive Function and TMS	Reviewed the effects of rTMS on motor cortical excitability and inhibition.

Attentional modulation: Attention plays a crucial role in selecting and amplifying relevant neural oscillations, shaping conscious content. Attention can be thought of as a spotlight that enhances the processing of specific information while suppressing irrelevant data. This selective process is mediated by the synchronization of neural oscillations in the attended regions. Research by Buschman and Miller (2007) demonstrated that attention modulates the synchronization of beta (15-30 Hz) and gamma oscillations in the prefrontal and parietal cortices, which are critical for top-down control of attention⁵.

Working memory: Working memory mechanisms actively maintain and manipulate conscious content through the sustained synchronization of neural oscillations. Working memory allows for the temporary storage and manipulation of information necessary for complex cognitive tasks. Studies have shown that theta (4-8 Hz) and gamma oscillations play a key role in maintaining information in working memory. Lisman and Jensen (2013) proposed that the interaction between theta and gamma oscillations supports the encoding and retrieval of information in working memory⁶.

Social context: Social interactions and cultural factors influence the development and modulation of neural oscillations, shaping the content and quality of conscious experience. Social context can affect the way individuals perceive and interpret their environment, which in turn influences neural oscillatory patterns. For example, research by Hari and Kujala (2009) suggested that social interactions can modulate neural oscillations in the mirror neuron system, which is involved in understanding and empathizing with others' actions and emotions⁷.

Implications for consciousness disorders

The DBH offers a potential framework for understanding various consciousness disorders, such as disorders of consciousness (DOC) and psychiatric disorders. Disruptions in neural oscillations, attention

or working memory may contribute to alterations in conscious experience in these conditions.

Disorders of Consciousness (DOC): In conditions like coma, vegetative state and minimally conscious state, disruptions in neural oscillations and global network connectivity may underlie impaired consciousness. The DBH suggests that targeting these neural mechanisms could potentially improve consciousness and cognitive function in DOC patients. For instance, studies have shown that patients in a vegetative state exhibit reduced gamma oscillatory activity and impaired connectivity between cortical regions³¹. Therapeutic interventions aimed at enhancing neural oscillatory activity and connectivity may hold promise for improving outcomes in DOC patients.

Psychiatric Disorders: Psychiatric disorders, such as schizophrenia and major depressive disorder, are often characterized by altered consciousness and cognitive impairments. The DBH provides a potential explanation for these symptoms, suggesting that disruptions in neural oscillations and attentional processes may contribute to these conditions. For example, schizophrenia has been associated with abnormalities in gamma oscillations and impaired synchronization between brain regions⁹. Similarly, major depressive disorder has been linked to altered alpha (8-12 Hz) and theta oscillations, which may reflect disruptions in attentional and working memory processes³².

FUTURE DIRECTIONS

To further develop and test the DBH, future research should focus on the following:

Neurophysiological studies: Employing advanced neuroimaging techniques (e.g., EEG, MEG, fMRI) to investigate the neural correlates of consciousness, particularly the role of neural oscillations and their synchronization. These studies can provide insights

into the temporal dynamics of neural oscillations and their role in conscious experience. For example, high-density EEG and MEG can be used to measure the precise timing and spatial distribution of neural oscillations, while fMRI can provide complementary information about the underlying neural networks³³.

Computational modeling: Developing computational models to simulate the dynamic binding of neural oscillations and explore the emergence of conscious experience. Computational models can help to test hypotheses about the mechanisms underlying neural oscillations and their role in consciousness. For instance, models of neural oscillatory networks can be used to investigate how different frequency bands interact to support cognitive processes and conscious experience³⁴.

Clinical applications: Applying the DBH to understand and treat consciousness disorders, such as DOC and psychiatric disorders. Clinical research can explore the potential of neuromodulation techniques (e.g., transcranial magnetic stimulation, transcranial direct current stimulation) to enhance neural oscillatory activity and improve cognitive function in patients with consciousness disorders. For example, transcranial magnetic stimulation (TMS) has been shown to modulate neural oscillations and improve cognitive function in patients with depression³⁵.

Translational research: Translating findings from basic research to clinical applications, developing novel therapeutic interventions for consciousness disorders. This involves bridging the gap between laboratory studies and clinical practice, ensuring that insights gained from neurophysiological and computational studies are effectively applied to improve patient outcomes. For instance, personalized neuromodulation protocols based on individual neural oscillatory profiles could be developed to target specific neural circuits involved in consciousness and cognitive function²⁷. According to the DBH, conscious experience arises from the dynamic integration of neural activity, specifically through synchronized neural oscillations. This integration is shaped by key components including neural oscillations themselves, attentional modulation, working memory, and social context. The interplay of these elements leads to conscious experience, and disruptions in this process are relevant to understanding both disorders of consciousness (DOC), such as coma and vegetative states, and psychiatric disorders like schizophrenia and depression. To further explore DBH, the further neurophysiological studies should be embraced, computational modeling and clinical applications as key areas of investigation. Ultimately, translational research is essential to connect basic research findings to clinical practice, advancing the understanding of consciousness and developing potential interventions for related disorders.

DBH and personalized medicine

Personalized medicine, or precision medicine, tailors healthcare to individual patients by considering factors like genetics, environment and lifestyle. This approach aims to move beyond a one-size-fits-all model, providing the right treatment to the right patient at the

right time. Key aspects of personalized medicine include using genetic information to understand disease susceptibility and treatment response, identifying biomarkers to predict disease progression, involving patients in their care and utilizing advanced diagnostics. DBH and personalized medicine intersect in several ways. By understanding how neural oscillations contribute to consciousness, researchers can identify specific neural circuits and oscillatory patterns disrupted in disorders of consciousness (DOC). Personalized medicine can then use this information to develop targeted therapies, such as neuromodulation techniques, tailored to individual patients. Genetic and biomarker research can identify individuals at higher risk for consciousness disorders and inform preventive strategies and personalized treatment plans. Advanced neuroimaging techniques, like fMRI, can be used to diagnose DOC more accurately and monitor treatment effectiveness³⁶⁻³⁹. Additionally, the patient-centered approach of personalized medicine aligns with the DBH's emphasis on social context and individual differences in shaping conscious experience.

DBH association with quantum biology

The DBH and quantum biology are two intriguing fields that, while seemingly distinct, share a common goal: understanding the complex mechanisms underlying consciousness. DBH focuses on the dynamic integration of neural activity through synchronized oscillations, emphasizing the roles of neural oscillations, attention, working memory and social context. Quantum biology, on the other hand, explores how quantum mechanical phenomena like entanglement and superposition influence biological processes, potentially impacting neural activity and information processing. A possible intersection between these two fields lies in the potential role of quantum coherence in neural oscillations. Quantum coherence could enhance the efficiency and speed of information processing in the brain, supporting the synchronization of neural activity that is essential for consciousness. Microtubules, structural components within neurons, might play a key role in both DBH and quantum biology. While DBH emphasizes their role in maintaining neural oscillations and network connectivity, quantum biology suggests that microtubules could support quantum coherence and entanglement. Additionally, quantum entanglement could facilitate the integration of information from different sensory modalities and cognitive processes, a crucial aspect of conscious experience⁴¹⁻⁴⁶. Biophotons, weak light emitted by biological systems, might also play a role in neural communication, potentially influencing neural oscillations and synchronization.

Criticisms and potential shortcomings

The DBH has garnered interest for its novel approach to understanding consciousness, but it also faces several criticisms and potential shortcomings.

Complexity and testability: The DBH involves intricate mechanisms of neural oscillations and their synchronization, which can be challenging to test empirically. The complexity of these interactions

makes it difficult to design experiments that can definitively confirm or refute the hypothesis¹.

Overemphasis on oscillations: Critics argue that the DBH may place too much emphasis on neural oscillations at the expense of other important neural processes. While oscillations are crucial, other factors like synaptic plasticity and neurotransmitter dynamics also play significant roles in consciousness².

Lack of direct evidence: Although there is indirect evidence supporting the role of neural oscillations in consciousness, direct evidence linking specific oscillatory patterns to conscious experience is still limited. More robust and direct experimental data are needed to validate the DBH³.

Integration with other theories: The DBH needs to be integrated with other existing theories of consciousness, such as the Global Workspace Theory and Integrated Information Theory. A comprehensive understanding of consciousness likely requires a synthesis of multiple theoretical perspectives.

Individual Differences: The DBH may not fully account for individual differences in conscious experience. Factors such as genetic variability, developmental history and environmental influences can affect neural oscillations and their role in consciousness^{4,9,16,33,48}. It will be important to compare various theories of consciousness, outlining their advantages and limitations particularly concerning the DBH⁴⁹⁻⁵³. Each theory offers valuable insights and has its own strengths and weaknesses. The DBH provides a unique perspective by emphasizing the role of neural oscillations and their synchronization in the integration of information across different brain regions.

There are various theories of consciousness that possess relationships to the DBH. The Global Workspace Theory (GWT) suggests that consciousness arises from the integration of information within a global workspace, similar to a spotlight in a theatre, effectively explaining conscious access and information integration, supported by evidence and models. However, GWT might not fully capture the subjective aspect of consciousness and could oversimplify neural complexity by focusing on a single workspace, while in relation to DBH, GWT similarly emphasizes information integration across brain regions and offers a framework for conscious access and attention, but it may oversimplify the dynamic and distributed neural processes that DBH addresses through synchronized oscillations. The Integrated Information Theory (IIT) proposes that consciousness corresponds to a system's capacity to integrate information, quantified as " ϕ " (Φ), providing a quantitative measure of consciousness and addressing subjective experience. Yet, IIT is computationally intensive, difficult to apply to large systems, and criticized for being abstract and potentially unfalsifiable. In relation to DBH, IIT's focus on information integration aligns with DBH's emphasis on neural oscillations and network connectivity, but its abstract and computationally intensive nature contrasts with DBH's mechanistic approach to neural dynamics. The Higher-Order Thought Theory (HOT) posits that a

mental state becomes conscious when it is the subject of a higher-order thought, explaining self-awareness and reflective consciousness and aligning with some neuropsychological findings. However, HOT heavily relies on higher-order thoughts, which might not be necessary for all conscious experiences, and lacks direct empirical support. When compared to DBH, HOT's explanation of self-awareness complements DBH's focus on attention and working memory, but its reliance on higher-order thoughts may not be essential for the dynamic integration processes that DBH emphasizes. Recurrent Processing Theory (RPT) suggests consciousness arises from recurrent processing within and between brain areas, highlighting feedback loops and supported by visual processing evidence. Nevertheless, RPT might not fully explain information integration across modalities and is limited to specific neural processing types. In the context of DBH, RPT's emphasis on feedback loops supports DBH's focus on dynamic neural interactions, but it may not fully capture the broad integration of information across sensory and cognitive processes like DBH. Finally, Predictive Coding Theory proposes that the brain constantly models the environment and minimizes prediction errors, offering a unifying framework for perception and action supported by broad evidence. Despite this, it primarily focuses on sensory processing and may not fully address higher cognitive functions, also being complex and computationally demanding⁴⁹⁻⁵³. In relation to DBH, Predictive Coding's focus on minimizing prediction errors aligns with DBH's emphasis on neural synchronization and integration, but it is more focused on sensory processing, whereas DBH offers a broader framework for understanding consciousness.

Understanding consciousness through the Dynamic Binding Hypothesis: Holistic overview

It should be remembered that the DBH offers a framework for understanding consciousness as a product of synchronized neural oscillations, which are rhythmic patterns of electrical activity in the brain that integrate sensory information into coherent experiences. This integration of brain activity, modulated by attentional modulation - the process of enhancing relevant information while filtering out the irrelevant - and working memory, a system for temporary information storage and manipulation, along with social context, the influence of social and cultural factors, is proposed as the basis of conscious experience, the subjective awareness of surroundings and thoughts. DBH suggests that disruptions in these synchronized oscillations may underlie disorders of consciousness (DOC), such as coma and vegetative states and also psychiatric disorders like schizophrenia and depression, which are mental health conditions affecting mood, thinking and behavior. The exploration of DBH involves neurophysiological studies using techniques like EEG, MEG and fMRI to study brain activity, alongside computational modeling to simulate complex neural systems and aims towards clinical applications for diagnosis and treatment, driven by translational research that bridges basic science and clinical practice. Understanding brain wave frequencies

like gamma oscillations (high-frequency for sensory integration), theta oscillations (lower-frequency for memory) and alpha oscillations (relaxed wakefulness) is crucial, as is the concept of synchronization, the brain's coordination of activity and top-down control, where higher brain areas influence lower-level processes. The mirror neuron system, activating during action and observation, and neuroimaging techniques like EEG (Electroencephalography), MEG (Magnetoencephalography), fMRI (Functional Magnetic Resonance Imaging), and Transcranial Magnetic Stimulation (TMS) for brain stimulation, are also important tools in this field. While insightful, DBH is not without limitations and should be considered alongside other theories to achieve a comprehensive understanding of consciousness^{4,7,9,16,33-35,47,54}.

Integrating insights from various perspectives may be the most effective approach to unraveling this complex phenomenon.

CONCLUSIONS

The DBH presents a strong framework for deciphering the neural basis of consciousness. By highlighting the dynamic interplay of brain activity, attention, working memory and social context, DBH provides a more holistic explanation of conscious experience compared to earlier theories. To fully validate DBH's predictions and explore its clinical utility, further investigation is essential. Integrating neurophysiological, computational and clinical methodologies will be vital for deepening our comprehension of consciousness and creating effective treatments for disorders affecting it. Combining DBH with quantum biology could lead to richer consciousness models that incorporate both classical and quantum mechanisms. This interdisciplinary direction opens new research paths, such as examining quantum effects within neural oscillations and how quantum coherence might influence cognitive functions. Understanding these combined mechanisms could significantly impact the treatment of consciousness and psychiatric disorders, potentially leading to innovative therapies that target both neural and quantum processes. Furthermore, merging DBH with personalized medicine allows researchers and clinicians to develop more tailored, effective and patient-focused treatments for consciousness disorders and other neurological conditions. This integrated approach holds the promise of significantly improving patient outcomes and enhancing our broader understanding of the brain and consciousness. Finally, examining DBH in conjunction with other consciousness theories is important to address and overcome any limitations inherent in the DBH framework.

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AUTHOR'S CONTRIBUTIONS

Eissa ME: conceived the idea, writing the manuscript, literature survey, formal analysis, critical review.

DATA AVAILABILITY

Upon request, the accompanying author can furnish the empirical data used to bolster the findings of the study.

CONFLICT OF INTEREST

None to declare.

REFERENCES

1. Koch C. What Is Consciousness? *Nature* 2018 May; 557(7704):S8–12.
<https://doi.org/10.1038/d41586-018-05097-x>
2. Neuroscience News. Unlocking the Mind: The neuroscience behind our conscious reality. *Neuroscience News* 2023.
3. Groundbreaking Study Affirms Quantum Basis for Consciousness: A Paradigm Shift in Understanding Human Nature. *SciTechDaily* 2024.
<https://doi.org/10.1523/ENEURO.0291-24.2024>
4. Fries P. A mechanism for cognitive dynamics: neuronal communication through neuronal coherence. *Trends Cogn Sci* 2005;9(10):474–480.
<http://doi.org/10.1016/j.tics.2005.08.011>
5. Buschman TJ, Miller EK. Top-down versus bottom-up control of attention in the prefrontal and posterior parietal cortices. *Science* 2007; 315(5820):1860–1862.
<http://doi.org/10.1126/science.1138071>
6. Lisman JE, Jensen O. The theta-gamma neural code. *Neuron* 2013; 77(6):1002–1016.
<http://doi.org/10.1016/j.neuron.2013.03.007>
7. Hari R, Kujala MV. Brain basis of human social interaction: from concepts to brain imaging. *Physiol Rev* 2009; 89(2):453–479.
<http://doi.org/10.1152/physrev.00041.2007>
8. Boly M, Tshibanda L, Vanhaudenhuyse A, et al. Functional connectivity in the default network during resting state is preserved in a vegetative but not in a brain dead patient. *Hum Brain Mapp* 2009;30(8):2393–2400.
<http://doi.org/10.1002/hbm.20672>
9. Uhlhaas PJ, Singer W. Abnormal neural oscillations and synchrony in schizophrenia. *Nat Rev Neurosci* 2010; 11(2):100–113. <http://doi.org/10.1038/nrn2774>
10. Klimesch W. Alpha-band oscillations, attention, and controlled access to stored information. *Trends Cogn Sci* 2012;16(12):606–617. <http://doi.org/10.1016/j.tics.2012.10>
11. Deco G, Jirsa VK, McIntosh AR. Emerging concepts for the dynamical organization of resting-state activity in the brain. *Nat Rev Neurosci* 2011; 12(1):43–56.
<http://doi.org/10.1038/nrn2961>
12. Fitzgerald PB, Fountain S, Daskalakis ZJ. A comprehensive review of the effects of rTMS on motor cortical excitability and inhibition. *Clin Neurophysiol* 2006; 117(12):2584–2596.
<http://doi.org/10.1016/j.clinph.2006.06.712>
13. Fries P, Reynolds JH, Rorie AE, Desimone R. Modulation of oscillatory neuronal synchronization by selective visual attention. *Science* 2001; 291(5508):1560–1563.
<http://doi.org/10.1126/science.1055465>
14. Engel AK, Fries P, Singer W. Dynamic predictions: oscillations and synchrony in top-down processing. *Nat Rev Neurosci* 2001;2(10):704–716.
<http://doi.org/10.1038/35094565>

15. Varela F, Lachaux JP, Rodriguez E, Martinerie J. The brainweb: phase synchronization and large-scale integration. *Nat Rev Neurosci* 2001;2(4):229-239. <http://doi.org/10.1038/35067550>
16. Buzsáki G, Draguhn A. Neuronal oscillations in cortical networks. *Science*. 2004; 304(5679):1926-1929. <http://doi.org/10.1126/science.1099745>
17. Siegel M, Donner TH, Engel AK. Spectral fingerprints of large-scale neuronal interactions. *Nat Rev Neurosci* 2012;13(2):121-134. <http://doi.org/10.1038/nrn3137>
18. Wang XJ. Neurophysiological and computational principles of cortical rhythms in cognition. *Physiol Rev* 2010; 90(3):1195-1268. <http://doi.org/10.1152/physrev.00035.2008>
19. Fell J, Axmacher N. The role of phase synchronization in memory processes. *Nat Rev Neurosci* 2011;12(2):105-118. <http://doi.org/10.1038/nrn2979>
20. Jensen O, Kaiser J, Lachaux JP. Human gamma-frequency oscillations associated with attention and memory. *Trends Neurosci* 2007;30(7):317-324. <http://doi.org/10.1016/j.tins.2007.05.001>
21. Fries P. Rhythms for cognition: communication through coherence. *Neuron* 2015; 88(1):220-235. <http://doi.org/10.1016/j.neuron.2015.09.034>
22. Buzsáki G. Rhythms of the Brain. Oxford University Press; 2006. <http://doi.org/10.1093/acprof:oso/9780195301069.001.0001>
23. Singer W. Neuronal synchrony: a versatile code for the definition of relations? *Neuron* 1999;24(1):49-65. [http://doi.org/10.1016/S0896-6273\(00\)80821-1](http://doi.org/10.1016/S0896-6273(00)80821-1)
24. Uhlhaas PJ, Roux F, Rodriguez E, Rotarska-Jagiela A, Singer W. Neural synchrony and the development of cortical networks. *Trends Cogn Sci* 2010; 14(2):72-80. <http://doi.org/10.1016/j.tics.2009.12.002>
25. Palva S, Palva JM. New vistas for alpha-frequency band oscillations. *Trends Neurosci* 2007;30(4):150-158. <http://doi.org/10.1016/j.tins.2007.02.001>
26. Siegel M, Warden MR, Miller EK. Phase-dependent neuronal coding of objects in short-term memory. *Proc Natl Acad Sci USA* 2009;106(50):21341-21346. <http://doi.org/10.1073/pnas.0908193106>
27. Thut G, Miniussi C. New insights into rhythmic brain activity from TMS-EEG studies. *Trends Cogn Sci* 2009;13(4):182-189. <http://doi.org/10.1016/j.tics.2009.01.004>
28. Canolty RT, Knight RT. The functional role of cross-frequency coupling. *Trends Cogn Sci* 2010; 14(11):506-515. <http://doi.org/10.1016/j.tics.2010.09.001>
29. Nani A, Manuella J, Mancuso L, Liloia D, Costa T, Cauda F. The Neural Correlates of Consciousness and Attention: Two Sister Processes of the Brain. *Front Neurosci* 2019 Oct 30; 13:1169. <http://doi.org/10.3389/fnins.2019.01169>
30. Valencia AL, Froese T. What binds us? Inter-brain neural synchronization and its implications for theories of human consciousness. *Neurosci Conscious* 2020 Jun 11;2020(1):niaa010. <http://doi.org/10.1093/nc/niia010>
31. Cruse D, Chennu S, Chatelle C, et al. Bedside detection of awareness in the vegetative state: A cohort study. *Lancet*. 2011 Nov 12; 378(9809):2088-94. [http://doi.org/10.1016/S0140-6736\(11\)61224-5](http://doi.org/10.1016/S0140-6736(11)61224-5)
32. Olbrich S, Arns M. EEG biomarkers in major depressive disorder: Discriminative power and prediction of treatment response. *Int Rev Psychiatry* 2013 Aug;25(5):604-18. <http://doi.org/10.3109/09540261.2013.816269>
33. Michel CM, Murray MM. Towards the utilization of EEG as a brain imaging tool. *Neuroimage* 2012 Aug 15;61(2):371-85. <http://doi.org/10.1016/j.neuroimage.2011.12.039>
34. Deco G, Jirsa VK, Robinson PA, Breakspear M, Friston K. The dynamic brain: from spiking neurons to neural masses and cortical fields. *PLoS Comput Biol* 2008 Jun 27;4(8):e1000092. <http://doi.org/10.1371/journal.pcbi.1000092>
35. Luber B, Lisanby SH. Enhancement of human cognitive performance using transcranial magnetic stimulation (TMS). *Neuroimage* 2014 Jan 15;85 Pt 3:961-70. <http://doi.org/10.1016/j.neuroimage.2013.06.007>
36. Collins FS, Varmus H. A new initiative on precision medicine. *N Engl J Med* 2015 Feb 26; 372(9):793-5. <http://doi.org/10.1056/NEJMp1500523>
37. Hamburg MA, Collins FS. The path to personalized medicine. *N Engl J Med*. 2010 Jul 22;363(4):301-4. <http://doi.org/10.1056/NEJMp1006304>
38. Hood L, Friend SH. Predictive, personalized, preventive, participatory (P4) cancer medicine. *Nat Rev Clin Oncol* 2011 Mar;8(3):184-7. <http://doi.org/10.1038/nrclinonc.2010.227>
39. Insel TR. Rethinking schizophrenia. *Nature*. 2010 Nov 11;468(7321):187-93. <http://doi.org/10.1038/nature09552>
40. Hameroff S, Penrose R. Consciousness in the universe: A review of the 'Orch OR' theory. *Phys Life Rev* 2014 Mar;11(1):39-78. <http://doi.org/10.1016/j.plrev.2013.08.002>
41. Rapp PE. Commentary on Wright and Bourke. The growth of cognition: Free energy minimization and the embryogenesis of cortical computation. *Physics of Life Reviews* 2021 Mar; 36:24-6. <http://doi.org/10.1016/j.plrev.2020.09.001>
42. Craddock TJ, Tuszynski JA, Hameroff S. Cytoskeletal signaling: Is memory encoded in microtubule lattices by CaMKII phosphorylation? *PLoS Comput Biol* 2012 Mar;8(3):e1002421. <http://doi.org/10.1371/journal.pcbi.1002421>
43. Popp FA, Gu Q, Li KH. Biophoton emission: experimental background and theoretical approaches. *Modern Physics Letters B* 1994 Sep 20;8(21n22):1269-96. <https://doi.org/10.1142/S0217984994001266>
44. Tuszynski JA. From quantum chemistry to quantum biology: a path toward consciousness. *Journal of Integrative Neuroscience* 2020 Dec 30;19(4):687-700. <https://doi.org/10.31083/j.jin.2020.04.393>
45. Hunter P. A quantum leap in biology: One inscrutable field helps another, as quantum physics unravels consciousness. *EMBO reports* 2006 Oct 1;7(10):971-4. <https://doi.org/10.1038/sj.embor.7400802>
46. Hameroff SR, Craddock TJ, Tuszynski JA. Quantum effects in the understanding of consciousness. *Journal of integrative neuroscience* 2014 Jun 9;13(02):229-52. <https://doi.org/10.1142/S0219635214400093>
47. Baddeley A. Working memory: theories, models, and controversies. *Annu Rev Psychol* 2012 Jan 10;63:1-29. <http://doi.org/10.1146/annurev-psych-120710-100422>
48. Dehaene S, Changeux JP. Experimental and theoretical approaches to conscious processing. *Neuron* 2011 Apr 28;70(2):200-27. <http://doi.org/10.1016/j.neuron.2011.03.018>
49. Baars BJ. The conscious access hypothesis: origins and recent evidence. *Trends Cogn Sci* 2002 Jan;6(1):47-52. [http://doi.org/10.1016/S1364-6613\(00\)01819-2](http://doi.org/10.1016/S1364-6613(00)01819-2)
50. Tononi G. An information integration theory of consciousness. *BMC Neurosci* 2004 Nov 3;5:42. <http://doi.org/10.1186/1471-2202-5-42>
51. Rosenthal D. Consciousness and Mind. Oxford University Press; 2005.
52. Lamme VA. Why visual attention and awareness are different. *Trends Cogn Sci* 2003 Jan;7(1):12-8. [http://doi.org/10.1016/S1364-6613\(02\)00013-X](http://doi.org/10.1016/S1364-6613(02)00013-X)
53. Friston K. The free-energy principle: a unified brain theory? *Nat Rev Neurosci* 2010 Feb; 11(2):127-38. <http://doi.org/10.1038/nrn2787>
54. Laureys S, Boly M, Maquet P. Tracking the recovery of consciousness from coma. *J Clin Invest* 2006 Jul; 116(7):1823-5. <http://doi.org/10.1172/JCI29172>